Waveform-Dependent Information Transfer in a Driven Kagome XY Model

Abstract

We investigate whether the temporal structure of an external drive, independent of its injected energy, shapes the response of a kagome XY lattice near its Berezinskii-Kosterlitz-Thouless transition. Using parametric bond driving $K(t) = K_0 + \delta K(t)$, we compare three canonical waveforms — sine, white noise, and random telegraph — at matched injected power.

Omnibus tests (one-way ANOVA and Kruskal-Wallis) show that the three waveforms produce statistically distinguishable absorbed energy, vortex-birth rate, and transfer entropy (TE) from input to birth (all $p<0.01$; Kruskal-Wallis $\varepsilon^2=0.71$–0.88). White noise absorbs the most energy yet carries no directional information: despite injecting the most energy of the three, it yields a null transfer-entropy result (net $TE=-0.0004\pm0.0016$). Telegraph and sine absorb statistically indistinguishable energy (Holm-corrected $p=0.15$) yet are clearly separated by transfer entropy ($p=0.024$) — energy alone cannot explain this distinction.

A square-wave period sweep, restricted to conditions satisfying a pre-registered sampling-adequacy protocol, reveals a near-linear internal relationship between absorbed energy and transfer entropy within that family; extrapolating it to sine's energy under-predicts the observed sine transfer entropy by roughly threefold, suggesting a waveform-family-dependent deviation from the internal square-wave scaling law. A complementary telegraph dwell-time sweep shows transfer entropy and overall effect size moving in opposite directions as the input's correlation time increases.

Together, these results demonstrate that temporal organization constitutes an independent control variable of nonequilibrium lattice dynamics, distinct from injected energy itself.

1. Introduction

The Berezinskii-Kosterlitz-Thouless (BKT) transition [1,2] describes the unbinding of topological defect pairs in two-dimensional systems without conventional symmetry breaking. The companion studies in this series established, first, that the kagome XY model's BKT transition leaves a distinct signature in the lattice's spatial information structure — a system-size-independent peak in the vortex-emergence efficiency $\eta = \text{birth}/\sqrt{(\text{MI})}$ (Paper I) — and second, that a local energetic descriptor (drop, the bond-energy excess around a vortex core) mediates an Energy → drop → Birth causal chain in equilibrium, while the collective information density $\sqrt{(\text{MI})}$ remains an emergent property that cannot be reduced to any single-vortex observable (Paper II).

Both prior studies examined the lattice at or near equilibrium. Neither asked a question that is, in some sense, prior to both: when energy is actively injected into the lattice, does the lattice merely respond to how much energy arrives, or does it also respond to how that energy is organized in time?

This question matters because two inputs can deliver identical average power while differing arbitrarily in their temporal structure — a sine wave delivers energy periodically and continuously; white noise delivers it as uncorrelated impulses; a random telegraph process delivers it as correlated but stochastic switching. If the lattice's response depends only on the amount of energy received, these three inputs should be indistinguishable once their power is matched. If the response depends on more than energy — for instance, on the input's frequency content, correlation time, or continuity — the three should be statistically separable.

We test this directly. Section 2 describes the driven kagome XY model, the three canonical input waveforms, two extended waveform families used to map out timescale dependence, and the statistical methodology, including a pre-registered sampling-adequacy protocol that guards against a temporal-aliasing artifact we identified during pilot analysis (Section 2.6). Section 3 presents the results. Section 4 discusses their interpretation and relation to Papers I and II. Section 5 concludes.

Figure 1 previews the argument developed in the remainder of this paper: a waveform's temporal structure sets how much microscopic disturbance can accumulate before it is lost to decorrelation, and this accumulated quantity, not absorbed energy, is what the lattice's response tracks. Absorbed energy and transfer entropy are therefore not interchangeable measures of a waveform's effect on the lattice, and identical injected power does not imply identical lattice response.

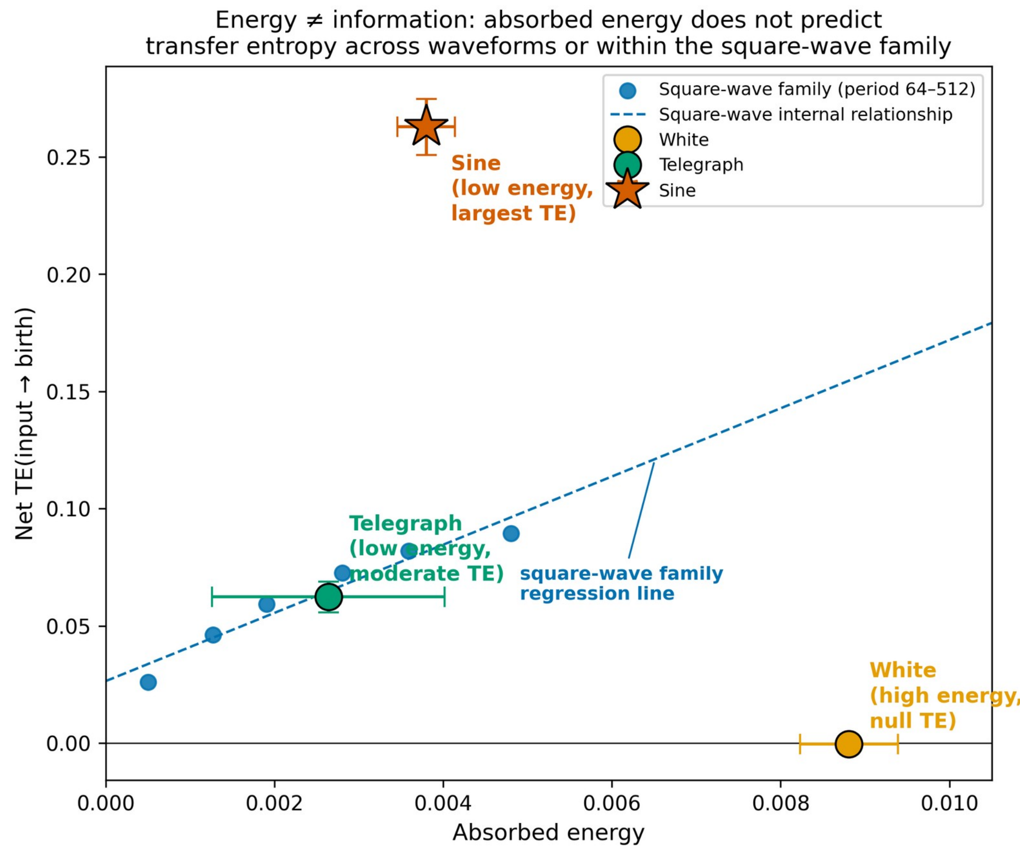

Figure 1. Waveform-dependent decoupling between absorbed energy and information transfer — the central discovery of this work. Although square-wave perturbations obey a nearly linear internal energy–TE relation (blue, $R^2 \approx 0.94$), the sinusoidal drive (star) lies far outside this family despite comparable absorbed energy to several square-wave periods, revealing a waveform-dependent discontinuity and demonstrating that temporal waveform itself constitutes an independent control parameter. White and Telegraph (matched-power canonical waveforms) are shown for reference; this pattern is established statistically in Sections 3.2–3.5.

This immediately raises a fundamental question: does the lattice merely absorb injected energy, or does it discriminate the temporal organization of the incoming perturbation? The remainder of this paper answers this question directly.

2. Model and Methods

Figure 2 outlines the overall measurement pipeline used throughout this study, from the external perturbation to the statistical comparison across waveforms.

Figure 2. Measurement pipeline. An external perturbation with a specified temporal structure (sine, white noise, or telegraph) drives the bond coupling K(t) of the kagome XY lattice; Monte Carlo evolution under this time-dependent Hamiltonian produces response observables (absorbed energy, vortex-birth rate, transfer entropy, phase lag), which are compared statistically across waveforms (Section 2.7).

2.1 Driven kagome XY model

We use the same kagome-lattice XY model as Papers I and II: $H(t) = -K(t)\,\Sigma\langle ij\rangle \cos(\varphi_i - \varphi_j)$, with $N=3L^2$ sites, L=48, evolved by the Metropolis algorithm. The lattice is held at fixed temperature T=0.9 and baseline coupling K_0=0.7, chosen to sit within the BKT critical region identified in Paper I. Rather than holding K fixed as in Papers I–II, we drive the bond coupling parametrically: $K(t) = K_0 + \delta K(t)$, where δK(t) is one of the input waveforms described below. This preserves the U(1) symmetry of the Hamiltonian while allowing the system to do external work, $W = \int(\partial H/\partial t)dt$, on the lattice.

2.2 Canonical input waveforms

These three signals were selected because they represent three qualitatively different temporal organizations while holding injected power fixed: (i) a memoryless stochastic input (white noise), (ii) a finite-memory stochastic input (random telegraph), and (iii) a deterministic periodic input (sine). Together they span the space of temporal correlation structure from none to bounded to unbounded, independent of signal amplitude.

Three canonical waveforms are compared at matched injected power $P = \langle\delta K^2\rangle = 0.01$:

Sine: $\delta K(t) = A \sin(\omega t + \varphi_0)$, with period_sweeps = 64 (chosen to satisfy the sampling-adequacy protocol of Section 2.6; see Section 3.5 for the consequence of an inadequately sampled period).

White noise: δK(t) drawn i.i.d. from a Gaussian distribution at every sweep, with a flat power spectrum.

Random telegraph: δK(t) switches between ±A with a geometric dwell-time distribution (mean dwell = 40 sweeps, likewise chosen to satisfy the sampling-adequacy protocol).

All three signals are normalized to the same injected power P before use (Table 1 of the Supplementary code documentation).

2.3 Extended waveform families

To map how the response depends on timescale, we additionally sweep two parametrized waveform families at fixed power:

Square-wave period sweep: a deterministic square wave δK(t) = ±A with period_sweeps ∈ {2,4,8,16,32,64,96,128,192,256,512}. This family generalizes the random telegraph's dwell=1 limit (a deterministic, period-2 alternation) into an independently tunable, fully deterministic waveform.

Telegraph dwell-time sweep: the random telegraph process with mean dwell ∈ {1,2,5,10,20,40,80} sweeps, holding input Shannon entropy fixed at ≈1 bit throughout (a telegraph signal has exactly two states regardless of dwell time), so that only the input's correlation time varies.

2.4 Response observables

For each run we record, at every measurement interval: the mean bond energy, the vortex-birth rate (fraction of plaquettes acquiring nonzero winding number per sweep), the local restoring strength drop (Paper II), the collective information density √(MI) (Paper I), and the helicity modulus. From these we compute: absorbed energy (driven-minus-baseline mean energy), phase lag at the drive frequency (for periodic inputs) or cross-correlation lag (for aperiodic inputs), and transfer entropy $TE(X \rightarrow Y) = H(Y_t|Y_{t-1}) - H(Y_t|Y_{t-1},X_{t-1})$, estimated with a 6-bin discretization at lag 1 (as in Paper II). Net TE = TE(input → birth) − TE(birth → input) quantifies the dominant direction of information flow.

2.5 Effect size

Statistical significance alone does not indicate the practical magnitude of a difference. For every observable we additionally report Hedges' g (a small-sample-corrected Cohen's d) comparing the baseline (undriven) and driven segments of the same run, using the conventional interpretive bands ($|g|<0.2$ negligible, 0.2–0.5 small, 0.5–0.8 medium, >0.8 large; Cohen, 1988) as a descriptive, not inferential, guide.

2.6 Sampling-adequacy protocol

During pilot analysis we found that net TE, evaluated at a fixed measurement stride, could change sign depending only on how finely the drive period was sampled — an aliasing artifact of the discrete-bin, lag-1 TE estimator, not a physical effect (reproduced directly: identical physical conditions gave net TE = +0.17 at a fine sampling stride and −0.10 at the stride used elsewhere in this study). To prevent this artifact from contaminating our conclusions, we adopt a three-tier sampling-adequacy protocol, evaluated before including any TE result in this paper's conclusions: (i) samples per drive period ≥ 10 (fail below 5, warning 5–10); (ii) number of complete drive cycles within the run ≥ 50 (fail below 20, warning 20–50); (iii) for aperiodic (telegraph) inputs, the measured signal autocorrelation time must exceed the measurement stride. A condition is classified HIGH only if all applicable criteria pass; transfer-entropy results are drawn only from HIGH-classified conditions. Phase lag and cross-correlation, which we confirmed empirically to be far less sensitive to this artifact, are reported for all conditions regardless of tier. The practical consequence of violating this protocol is documented directly in the Appendix.

2.7 Statistical testing

For the three-way comparison among canonical waveforms (n=5 seeds per waveform), we report both one-way ANOVA and the non-parametric Kruskal-Wallis test (more appropriate given the small per-group sample size and no assumption of normality), together with the Kruskal-Wallis effect size ε^2. Where the omnibus test is significant, pairwise comparisons use the Mann-Whitney U test with Holm-Bonferroni correction for three comparisons.

3. Results

As anticipated by the global comparison shown in Figure 1, we now examine each component of this pattern separately and establish it statistically. White noise absorbs the most energy yet produces the lowest transfer entropy of the three; sine absorbs less energy than white noise yet produces by far the highest transfer entropy.

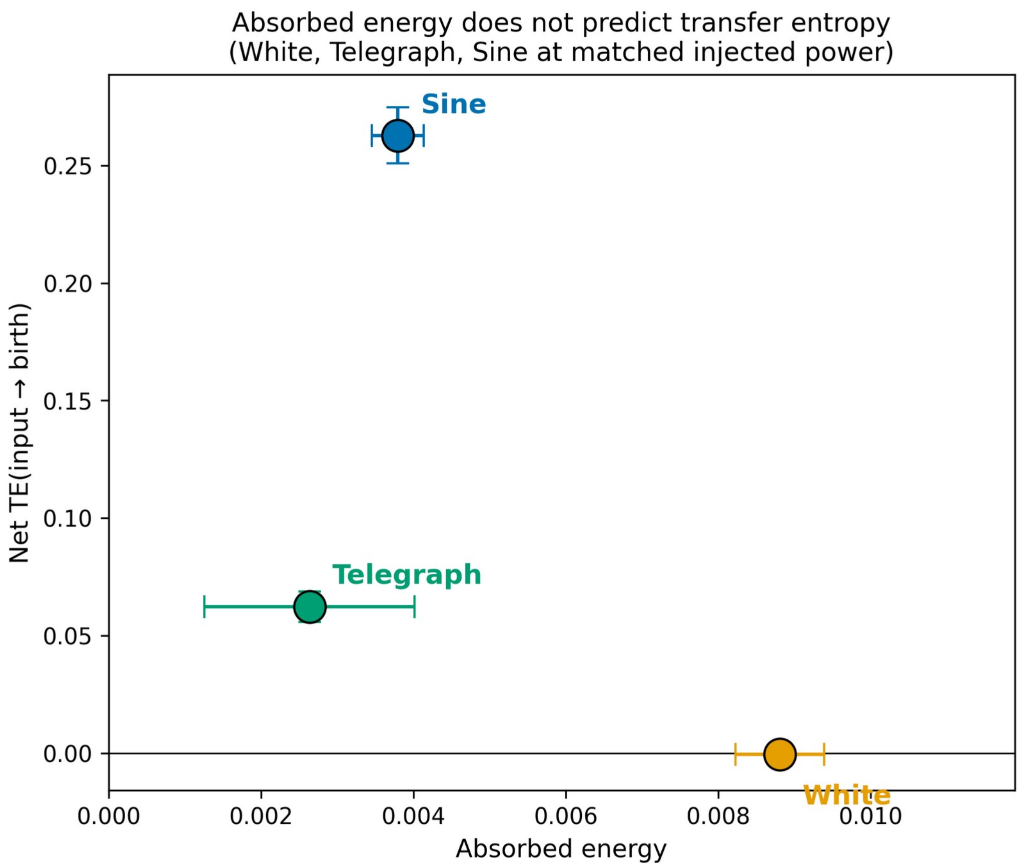

Figure 3. Absorbed energy versus net transfer entropy for the three canonical waveforms at matched injected power (mean ± 95% CI, n=5 seeds). Energy and transfer entropy are not monotonically related across waveforms.

Figure 3 summarizes each waveform by its mean and 95% CI; Figure 4 shows the same data at the level of individual seeds, to make the within-waveform spread and between-waveform separation directly visible.

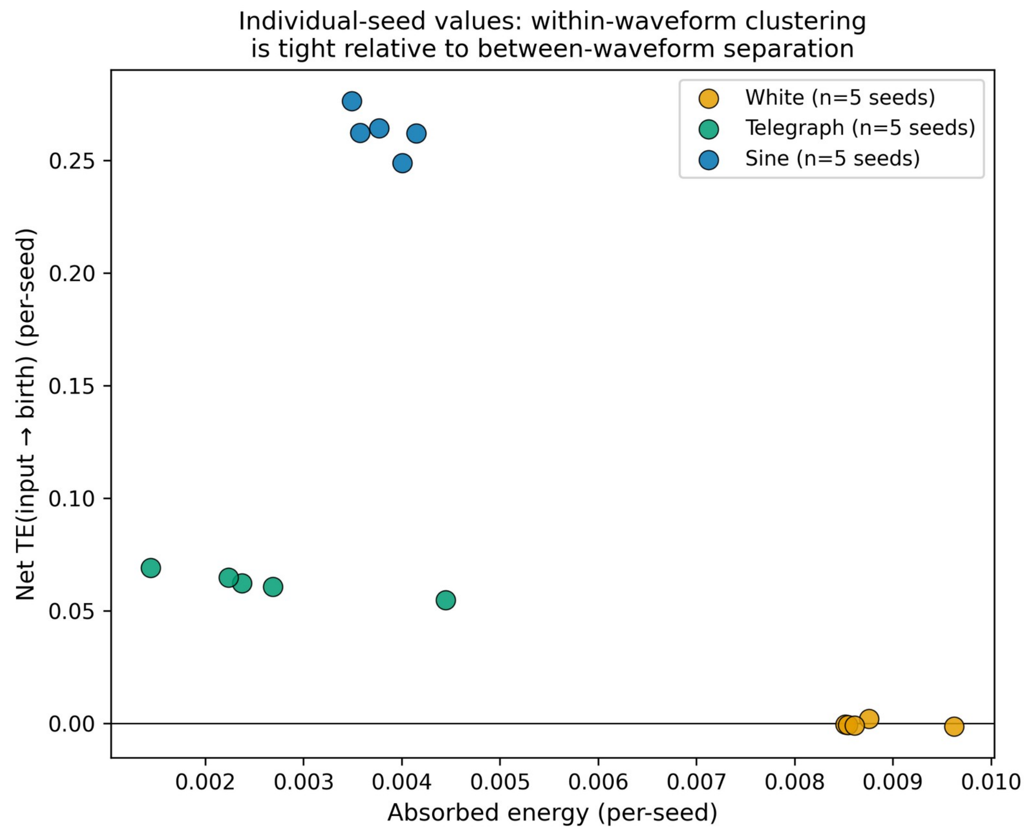

Figure 4. Individual-seed absorbed energy and net transfer entropy for all 15 runs (5 seeds × 3 waveforms). Within-waveform spread is small relative to between-waveform separation, particularly for transfer entropy.

3.1 The three canonical waveforms produce statistically distinct responses

Table 1 summarizes the omnibus tests across White, Telegraph, and Sine (5 seeds each, all HIGH-tier). Net TE, absorbed energy, Δbirth, and effect size on birth are all highly significant by both ANOVA and Kruskal-Wallis, with Kruskal-Wallis ε^2 in the 0.71–0.88 range — conventionally a very large effect.

Table 1. Omnibus tests across White, Telegraph, and Sine (n=5 seeds per waveform, all HIGH sampling-adequacy tier).

Response variable	ANOVA F (p)	Kruskal-Wallis H (p)	ε^2
Net TE(input → birth)	2294 (3×10^{-16})	12.5 (0.0019)	0.875
ΔBirth	31.8 (1.6×10^{-5})	10.5 (0.0052)	0.708
Absorbed energy	106 (2.4×10^{-8})	10.5 (0.0052)	0.708
Effect size (birth)	231 (2.6×10^{-10})	10.5 (0.0052)	0.708

3.2 Question 1: can absorbed energy alone explain the response?

White noise absorbs the most energy of the three canonical waveforms (0.00881±0.00058, also yielding the largest baseline-vs-driven effect size on birth) — yet its net transfer entropy is indistinguishable from zero (−0.0004±0.0016, HIGH tier). The lattice's birth rate responds substantially to white-noise driving in a simple before/after sense, but the temporal pattern of that response carries no statistically detectable directional information relative to the input. This null result is, in the present context, as informative as a positive one: injecting more energy is not equivalent to transmitting more information.

A second, independent test addresses the same question from the opposite direction. If energy alone governed the response, two waveforms absorbing statistically indistinguishable energy should be indistinguishable in their response more generally. Telegraph (dwell=40) and sine (period=64) absorb statistically indistinguishable energy (0.00264±0.00138 vs. 0.00380±0.00034; Mann-Whitney, Holm-corrected p=0.15). Under the energy hypothesis, no further difference between them should be observable; Section 3.3 tests this directly.

3.3 Question 2: does temporal organization succeed where energy fails?

The net transfer entropies of telegraph and sine are clearly separated (0.0623±0.0065 vs. 0.2628±0.0120; all three pairwise comparisons among White, Telegraph, and Sine reach p=0.024 after Holm correction), despite their statistically indistinguishable energy. The telegraph and sine waveforms therefore provide a controlled counterexample to an energy-only description: two inputs with statistically indistinguishable absorbed energy nevertheless produce measurably different directional information transfer. Figure 5 — the central result of this study — displays this pattern directly: energy fails to distinguish these two waveforms, while transfer entropy succeeds. Because transfer entropy is sensitive to the temporal order and correlation structure of the driven and response time series, not merely their means, this result identifies temporal organization, rather than energy magnitude, as the variable governing the lattice's waveform-selective information processing.

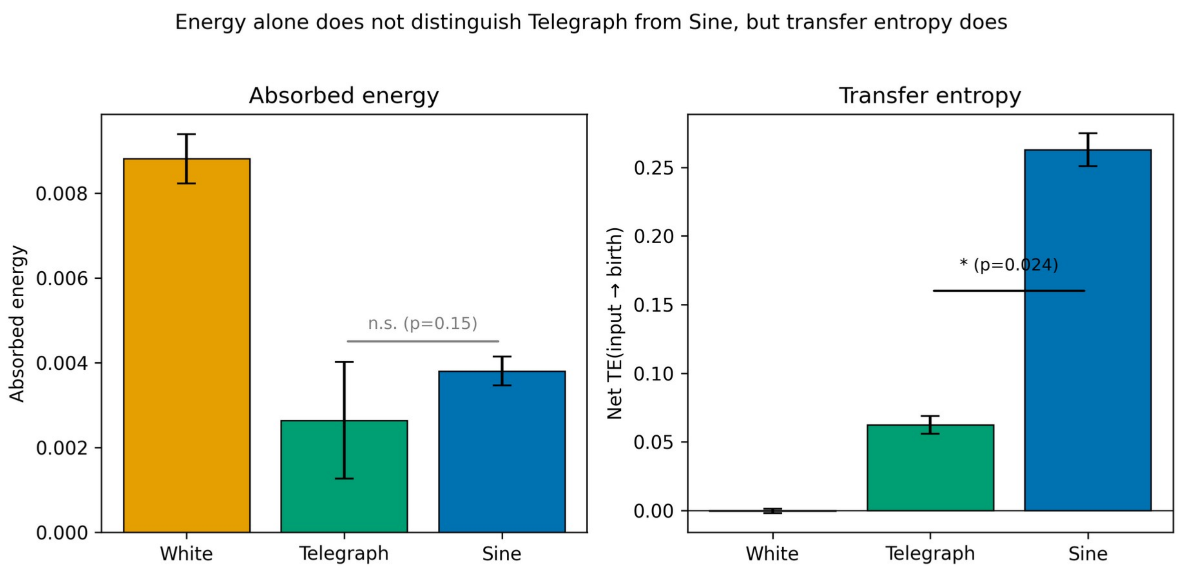

Figure 5. Energy alone does not distinguish Telegraph from Sine (left; Mann-Whitney p=0.15, n.s.), but transfer entropy does (right; p=0.024, Holm-corrected). Bars show mean ± 95% CI across 5 seeds.

The same result can be seen from a different angle: ranking the three waveforms by absorbed energy and by transfer entropy produces opposite orderings (Figure 6). White ranks first in energy and last in transfer entropy; sine ranks last in energy among the two driven-response comparisons of Section 3.2 and first in transfer entropy. No monotonic transformation of energy reproduces the transfer-entropy ranking.

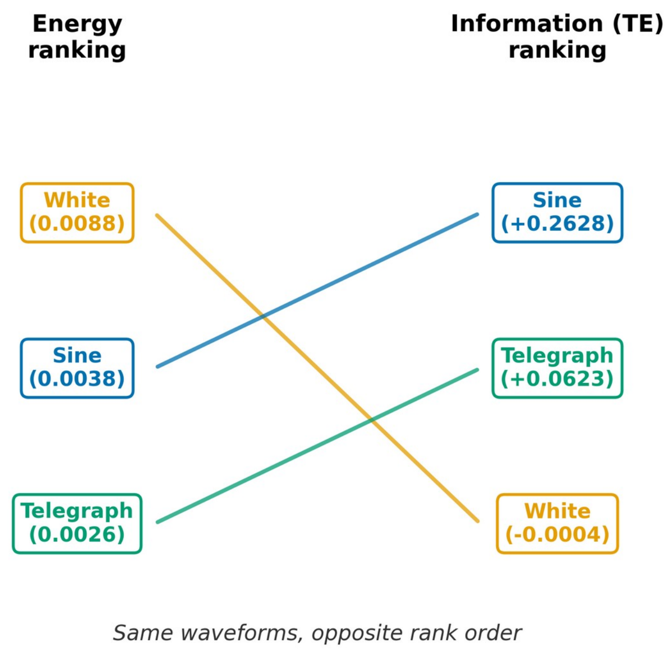

Figure 6. Ranking the three canonical waveforms by absorbed energy (left) versus by net transfer entropy (right) produces opposite orderings, illustrated by the crossing connector lines. Values in parentheses are absorbed energy (left) and net TE (right).

3.4 Question 3: does a single internal law hold within a waveform family?

Having shown that waveform identity matters, we ask whether the response within a single, well-controlled waveform family follows a predictable internal law. Within the square-wave family (periods 64–512, all HIGH tier), absorbed energy and net TE both decrease monotonically with period, and are related by a near-linear law (Pearson r=0.968): net TE increases by approximately 14.6 units per unit of absorbed energy within this waveform family. Absorbed energy falls by a factor of ≈9.6 from period=64 to period=512, while net TE falls by only a factor of ≈3.4 over the same range, so the transfer-entropy-per-unit-energy ratio rises from ≈18.6 to ≈51.8 — slower driving is a more efficient converter of absorbed energy into directional information transfer, within this waveform family.

3.5 Question 4: does this internal law generalize across waveform families?

Extrapolating the square-wave law of Section 3.4 to sine's absorbed energy (0.00380) predicts TE≈0.108; the observed sine TE (0.263, at the matched period=64) is approximately 3× higher (Figure 7). The answer is no, at least between the two families compared here: the square-wave family's internal energy-TE law does not extend to sine. Because the comparison uses sine and square wave at the identical period (64) and identical injected power, this discrepancy cannot be attributed to a difference in drive frequency or energy. With only two waveform families compared at this level of detail, we describe this as a waveform-family-dependent deviation from the internal square-wave scaling law, rather than as an established general phenomenon; further waveform families would be needed to determine how widely it generalizes.

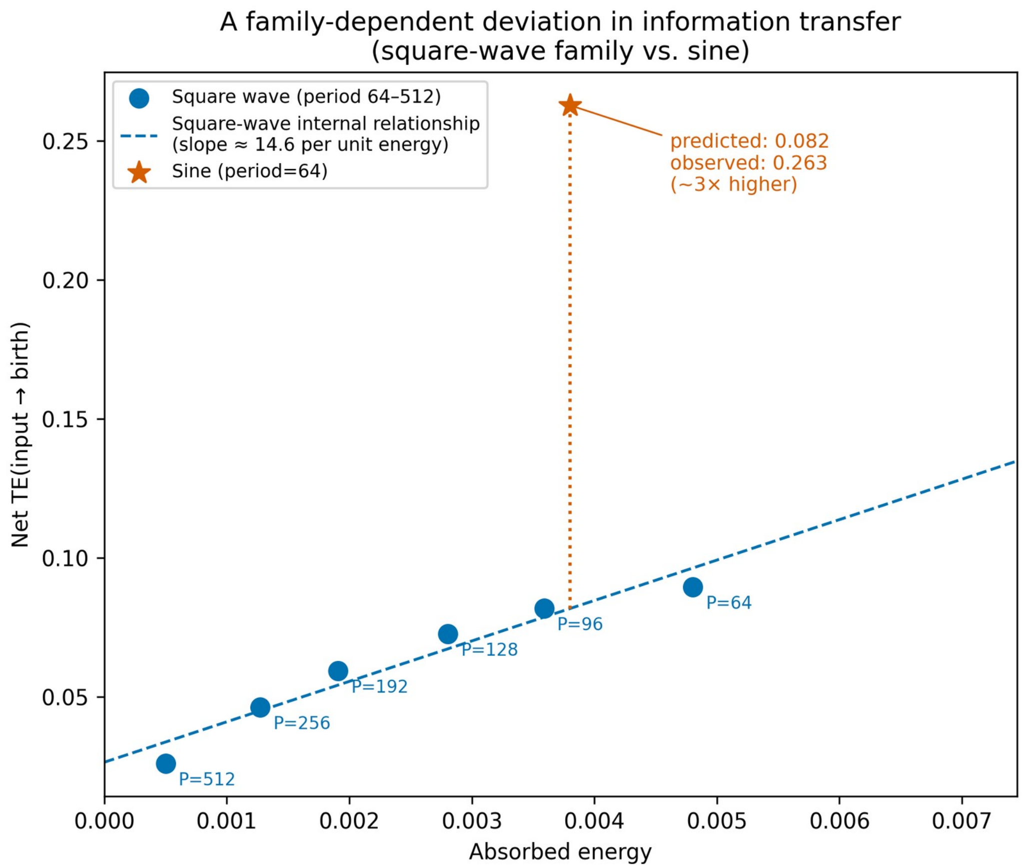

Figure 7. Failure of the square-wave energy–transfer-entropy law to predict the sine response. Square-wave period sweep establishes an internal linear relationship between absorbed energy and net TE within that waveform family (dashed line); sine at the same period and matched power (star) lies approximately 3× above the value this relationship predicts at its energy — a waveform-family-dependent deviation from this scaling law, observed here between two waveform families.

Figure 8 summarizes the two input axes examined across Sections 3.1–3.5. Energy sets a floor on the magnitude of the lattice's response; temporal organization is a separate axis that determines the directional information content of that response and, through it, the topological outcome.

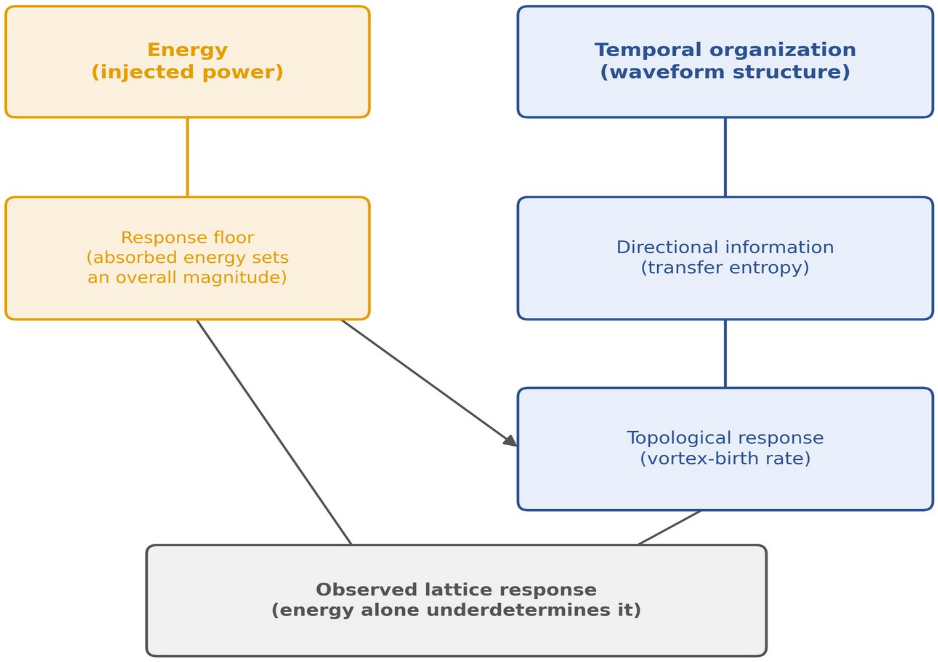

Figure 8. Two independent input axes shape the lattice's response. Energy sets a floor on the overall response magnitude; temporal organization separately determines the directional information content (transfer entropy) and the resulting topological outcome (vortex-birth rate). Energy alone underdetermines the observed response.

The results presented above collectively answer the central question posed in the Introduction. The lattice clearly distinguishes temporal organization even when injected energy alone does not.

These observations naturally raise a further question: if the response depends on accumulated temporal information rather than instantaneous energy, can the accumulated state also be actively erased?

4. Discussion

The principal finding of this work is not that different waveforms inject different amounts of energy, but that identical-power inputs generate statistically distinguishable lattice responses depending on their temporal organization: the lattice exhibits waveform-selective information processing. The present results therefore indicate that lattice dynamics discriminate temporal organization itself, rather than merely integrated injected energy. The temporal organization of the external perturbation constitutes an independent control parameter governing information transfer into the lattice. Thus, identical injected power does not imply identical lattice response.

4.1 Energy is necessary but not sufficient

Every result in Section 3 is consistent with a single picture: absorbed energy sets a floor on the lattice's response (white noise, with the most energy, produces the largest simple effect size), but does not determine the response's informational content. White noise's null transfer entropy, despite leading in both energy and effect size, and the telegraph/sine pair's matched energy but separated transfer entropy (Section 3.2), both point to the same conclusion from different directions. Among all response observables examined here, transfer entropy is the only quantity that statistically separates telegraph and sine driving despite their indistinguishable absorbed energies; we do not claim this generalizes beyond the conditions tested here.

4.2 A physical interpretation: temporal coherence of the drive

The results of Section 3 point toward a mechanistic picture beyond the purely phenomenological statement that waveform identity matters. We propose, as an interpretation rather than a directly measured mechanism, that the lattice does not respond to injected energy per se, but to the temporal coherence of the perturbation. A bond perturbation does not produce a vortex instantaneously; rather, it creates a sequence of local energetic rearrangements whose cumulative effect must persist over multiple Monte Carlo updates before a topological event becomes statistically probable. Coherently driven fluctuations persist long enough for this accumulation to occur, whereas incoherent, rapidly decorrelating fluctuations dissipate first.

This picture is consistent with the telegraph dwell-time sweep (Supplementary Note), where net transfer entropy tracks the input's correlation time rather than its fixed entropy, and it places White, Telegraph, and Sine at increasing points along a single accumulation axis: white noise's immediate decorrelation prevents accumulation entirely, telegraph's finite dwell time permits partial accumulation, and sine's sustained coherence permits accumulation over the longest uninterrupted interval. This picture also yields a directly testable prediction: reversing the drive's phase mid-experiment (counter-driving) should interrupt this accumulation and suppress the resulting transfer entropy and birth rate, whereas simply withdrawing the drive should not.

Figure 9 illustrates this proposed accumulation pathway schematically for each canonical waveform: a bond fluctuation, its persistence (or lack thereof), the resulting degree of microscopic accumulation, and the associated transfer entropy.

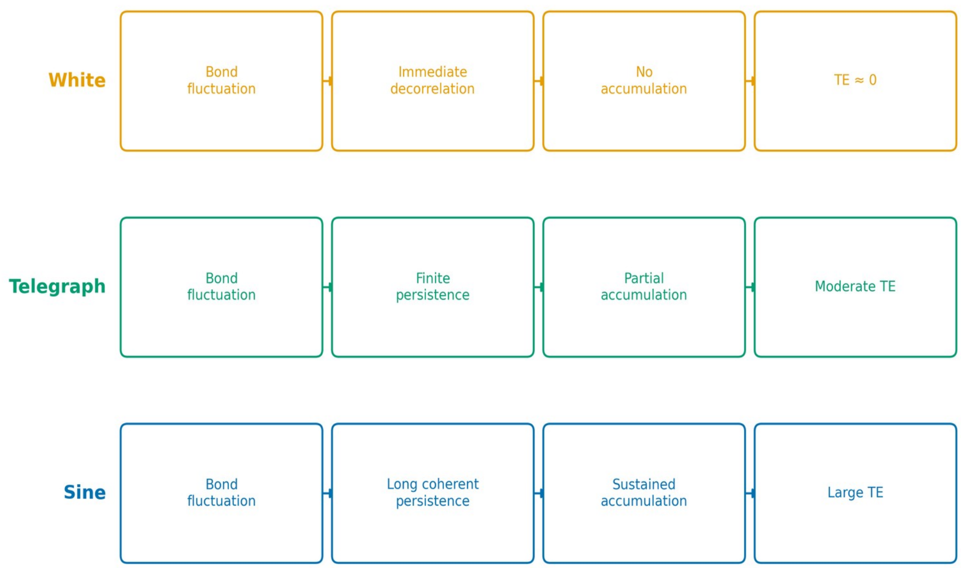

Figure 9. Proposed accumulation pathway from bond fluctuation to transfer entropy, shown schematically for each canonical waveform: temporal coherence sets how long a fluctuation persists, which in turn sets how much microscopic rearrangement can accumulate before a topological event becomes probable. This diagram illustrates an interpretation (Section 4.2), not a directly measured mechanism; the persistence and accumulation stages are not independently observed in this study.

We emphasize the limits of this interpretation. The data measure macroscopic consequences of temporal coherence (absorbed energy, birth rate, transfer entropy); they do not identify which internal lattice degree of freedom, if any, stores a memory of the drive's temporal structure. Candidate mechanisms — the spatial extent of the information field (√(MI)), its correlation length, or local topological quantities such as winding or chirality — are plausible but unmeasured in this study, and asserting any one of them here would let interpretation outrun the data. We reserve this question, and the morphological analysis needed to address it, for a dedicated follow-up study (Section 4.3).

4.3 Scope of this study

This study characterizes when and how strongly the lattice's macroscopic response (energy, birth rate, transfer entropy) depends on waveform identity; it does not address which internal spatial reorganization of the information field accompanies a given waveform's response. Morphological descriptors of the information field (spatial mean, maximum, standard deviation, skewness, connected-component structure) and raw field snapshots were recorded throughout this study's mining but are not analyzed here.

4.4 The waveform-family-dependent deviation

The approximately 3-fold discrepancy between sine and the square-wave family's internal law, at matched period and power (Section 3.4), is, to our knowledge, a new observation for this system. We intentionally avoid attributing the deviation to any single waveform property, because several properties (temporal continuity, harmonic spectrum, switching discontinuity) change simultaneously between sine and square-wave driving, and the present data do not isolate which is responsible. We therefore interpret Figure 7 only as evidence that waveform family itself constitutes an independent control parameter beyond absorbed energy, rather than as evidence for any single mechanism. A natural next step is to interpolate continuously between square and sine (e.g., via a tunable rise-time or a Fourier truncation series) to determine which property, if any, is responsible, and whether the deviation recurs across other waveform pairs.

One possible interpretation, offered here as a hypothesis rather than a conclusion, is that temporally smooth inputs preserve coherent lattice rearrangements over multiple Monte Carlo updates, whereas temporally discontinuous inputs repeatedly disrupt such correlations before they can propagate into a directional signal detectable by transfer entropy. This would predict that transfer entropy should increase smoothly as a waveform's discontinuity is reduced, independent of its energy content — a prediction directly testable with the continuity-interpolation experiment proposed above.

4.5 A methodological contribution

This result highlights a broader methodological point, first identified during pilot analysis for this paper and independently confirmed in the companion equilibrium analysis of Paper II: transfer-entropy analyses are sensitive to the relationship between the observation stride and the input's intrinsic timescale, and can produce sign-reversed or otherwise spurious directional conclusions when this relationship is inadequate, independent of any genuine underlying effect. We suggest that studies applying transfer entropy to driven or non-equilibrium systems adopt an explicit, pre-registered sampling-adequacy criterion of the kind used here (Section 2.6) before drawing directional conclusions — concretely, $\geq$10 samples per drive period and $\geq$50 complete drive cycles, which together prevented the sign-reversal artifact documented in Appendix Figure A1 from contaminating this paper's conclusions. We recommend reporting explicit sampling-adequacy diagnostics whenever transfer entropy is used for driven nonequilibrium systems, since inadequate sampling alone can reverse the inferred information-flow direction.

4.6 Relation to previous work

Papers I and II characterized the lattice's equilibrium information structure and its local energetic mechanism, respectively. Paper III adds a third axis: the lattice's response is not fully specified by its equilibrium state and the amount of energy subsequently injected into it. This is consistent with, and extends, Paper II's finding that energetic and informational descriptions of this system are not interchangeable. Unlike Papers I and II, which characterize what the lattice's information structure is, Paper III does not merely measure information — it is, to our knowledge, the first experiment on this system to show that the lattice actively discriminates the temporal organization of an external drive.

The present results establish that temporal information structure represents a control axis independent of injected energy. This provides an experimentally testable framework for studying information-driven lattice dynamics beyond conventional energetic descriptions.

Although demonstrated here for a driven kagome XY lattice, this picture suggests a framework that may extend to other systems in which collective topological rearrangements emerge from temporally accumulated microscopic fluctuations.

5. Conclusions

1. At matched injected power, sine, white noise, and random telegraph driving produce statistically distinguishable response dynamics in absorbed energy, vortex-birth rate, and transfer entropy (one-way ANOVA and Kruskal-Wallis, all $p<0.01$; Kruskal-Wallis $\varepsilon^2=0.71$–0.88, confirmed by Holm-corrected pairwise tests).
2. Energy alone does not account for this waveform dependence: white noise produces the largest absorbed energy and effect size yet a net transfer entropy indistinguishable from zero, while telegraph and sine — statistically indistinguishable in absorbed energy and effect size ($p=0.15$) — are nonetheless clearly separated by transfer entropy ($p=0.024$).
3. A square-wave period sweep (periods 64–512) establishes a near-linear internal relationship between absorbed energy and transfer entropy within that waveform family; sine, at matched period and power, departs from this relationship by a factor of ≈3 — a waveform-family-dependent deviation from this scaling law that warrants further investigation across additional waveform families.

Collectively, these observations imply that the lattice response cannot be characterized solely by energy absorption. Instead, temporal organization of the external perturbation constitutes an independent control parameter governing information flow into the lattice, producing statistically distinguishable, waveform-selective dynamics even under matched power injection. More broadly, these results suggest that driven lattice systems can be investigated from the perspective of information dynamics, in addition to conventional energetic response. These findings establish temporal information as an experimentally distinguishable control variable of lattice dynamics. This shifts the description of driven lattice dynamics from an energy-only viewpoint toward one in which temporal organization constitutes an independent, experimentally controllable variable.

Appendix: Sampling-adequacy diagnostics

Figure A1 shows why the sampling-adequacy protocol of Section 2.6 selects period≥64 for transfer-entropy conclusions: below this threshold, samples per period fall below the pre-registered PASS criterion (≥10).

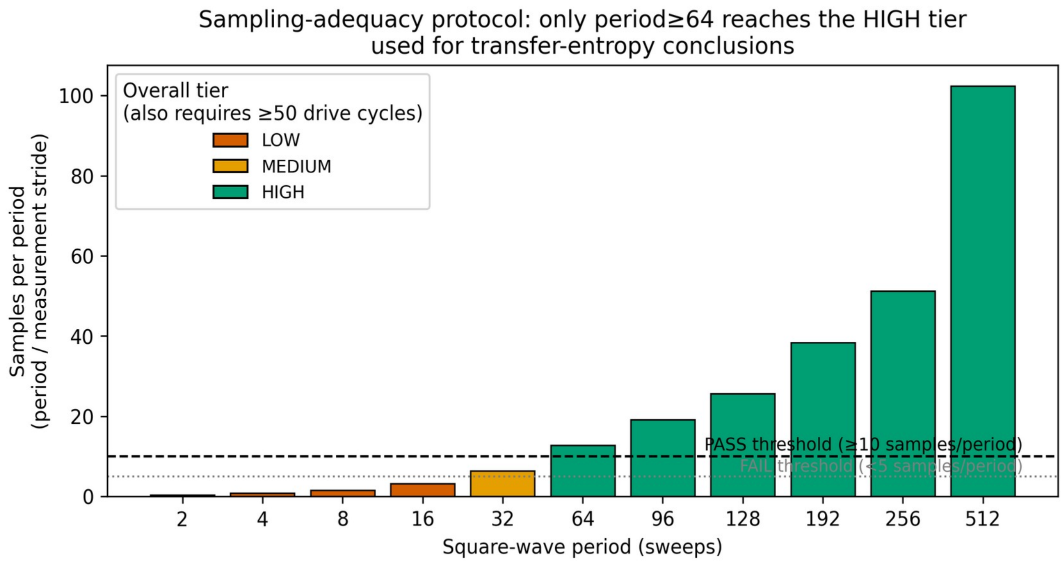

Figure A1. Samples per drive period, by square-wave period, at the measurement stride used throughout this study (5 sweeps). Color indicates the overall sampling-adequacy tier (also incorporating the ≥50-cycle criterion of Section 2.6); only HIGH-tier periods (≥64) are used for transfer-entropy conclusions in the main text.

Square-wave periods 2–32 fail the sampling-adequacy protocol of Section 2.6 at the measurement stride used in this study. Their net TE values are erratic in sign (e.g., period=16: net TE=−0.20; period=4: net TE=−0.04) and are excluded from the conclusions above; only phase lag, cross-correlation, and effect size — which are far less sensitive to this artifact — are considered reliable for these conditions. We report this explicitly because it is methodologically important: naively including these values would have suggested a non-monotonic or sign-changing energy-TE relationship where in fact the underlying relationship (Section 3.4) is smooth and monotonic once adequately sampled conditions are used.

Supplementary Note: Telegraph dwell-time sweep

As a supplementary check on the energy-information decoupling reported in Section 3.2, we also swept the random telegraph process's dwell time (input Shannon entropy held fixed at ≈1 bit throughout), independent of the canonical dwell=40 condition used elsewhere in this study. Net TE rises from ≈0.002 (dwell=2) to a plateau near 0.05–0.06 (dwell≥20), while the baseline-vs-driven effect size on birth falls monotonically from 0.420 to 0.109 over the same range — the two metrics move in opposite directions as the drive's correlation time increases, consistent with the energy/information decoupling that is this paper's central theme. As an internal consistency check, dwell=1 (a deterministic period-2 square wave, not a genuine stochastic process) gives an effect size (0.626) matching the independently measured square-wave period=2 result (0.615), from a separately mined dataset. This sweep supports, rather than drives, the paper's central claim and is reported here for completeness.

Acknowledgments

This work was carried out independently, without institutional affiliation. Large language models (Claude, ChatGPT) were used as auxiliary tools. All simulation design, data interpretation, and conclusions reflect the author's own judgment.

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